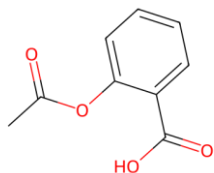


SMILES



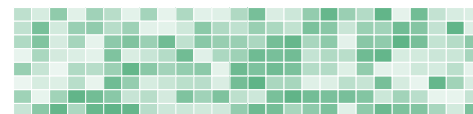
CC(=O)Oc1ccccc1C(=O)O

① 1D · Sequence

BPE Tokenizer
(AutoTokenizer)

ChemBERTa-77M-
MTR
(frozen)

1D Semantic Tokens



Semantic tokens [354 × 128]

special tokens removed · adaptive-pooled

② 2D · Graph

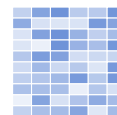
RDKit parse
atoms · bonds

Canonical
atom/bond
featurizer → DGL
graph

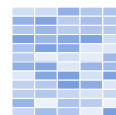
2D Molecular Graph



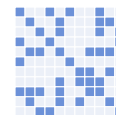
Molecular graph
nodes + edges



Node feats X
[290 × 75]



Edge feats E
[E × 12]



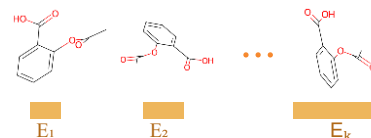
Adjacency A
[290 × 290]

③ 3D · Conformers

AddHs + ETKDGv3
embed K confs

UFF optimize
+ energy

K Conformers + Energy



K conformers(sorted by energy)

atom feats [K×64×128]

coords [K×64×3] (x,y,z)

energy [K] UFF → z-score

conf_mask [K] valid confs

X = 74 atom feats + 1 virtual-node bit
E = bond type · conjugation · ring · stereo A = self-loops on diagonal